

CX4240 Computing for Data Analysis - Homework 1

Name:
GTID:

Deadline: 11:59 pm EST, Feb 04

- Submit your answers as one single PDF file on Gradescope.
- You will be allowed 2 total late days (48 hours) without penalty for the entire semester. Once those days are used, you will be penalized according to the following policy:
 - Homework is worth full credit before the due time.
 - It is worth 75% credit for the next 24 hours.
 - It is worth 50% credit for the second 24 hours.
 - It is worth zero credit after that.
- You are required to use Latex, or word processing software, to generate your solutions to the written questions. Handwritten solutions WILL NOT BE ACCEPTED.

1 Linear Algebra [35pts]

Let $A \in \mathbb{R}^{n \times n}$ be a symmetric matrix. We say that A is **positive semidefinite (PSD)** if

$$x^T A x \geq 0 \quad \text{for all } x \in \mathbb{R}^n.$$

- (a) [10 points] Prove that if A is PSD, then cA is PSD for every scalar $c \geq 0$.
- (b) [10 points] Suppose A has an orthonormal eigenbasis $\{v_1, \dots, v_n\}$ with eigenvalues $\lambda_1, \dots, \lambda_n$. Show that for any $x \in \mathbb{R}^n$,

$$x^T A x = \sum_{i=1}^n \lambda_i (v_i^T x)^2.$$

- (c) [15 points] Use part (b) to prove the following eigenvalue criterion:

$$A \text{ is PSD} \iff \lambda_i \geq 0 \quad \text{for all } i = 1, \dots, n.$$

(Hint: You may use without proof that symmetric matrices have an orthonormal eigenbasis.)

Solution:

2 Probability and Statistics [35pts]

1. [15 points] A discrete random variable X takes values in $\{0, 1, 2, 3\}$ with

$$\mathbb{P}(X = 0) = 0.1, \quad \mathbb{P}(X = 1) = 0.2, \quad \mathbb{P}(X = 2) = 0.3, \quad \mathbb{P}(X = 3) = 0.4.$$

Let $X_1, \dots, X_5$ be i.i.d. copies of X , and define the sample mean

$$\bar{X} = \frac{1}{5} \sum_{i=1}^5 X_i.$$

Compute:

- (a) $\mathbb{E}[X]$ and $\text{Var}(X)$;
- (b) $\mathbb{E}[\bar{X}]$ and $\text{Var}(\bar{X})$.
- (c) Define $Y = 2X - 1$. Compute $\mathbb{E}[Y]$ and $\text{Var}(Y)$.

Solution:

2. [20 points]

Let $X \sim \mathcal{N}(\mu, \Sigma)$ and consider the affine transformation

$$Y = AX + b,$$

where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.

- (a) Derive the distribution of Y by computing its mean and covariance matrix.
- (b) Now let's consider the one-dimensional case and let $X \sim \mathcal{N}(\mu_1, \sigma_1)$ and $X' \sim \mathcal{N}(\mu_2, \sigma_2)$, where $X, X' \in \mathbb{R}$. The Kullback-Leibler (KL) divergence between two distributions with densities $p(x)$ and $q(x)$ is defined as

$$D_{\text{KL}}(p \parallel q) = \int_{\mathbb{R}^n} p(x) \log \frac{p(x)}{q(x)} dx.$$

Derive the closed-form expression for

$$D_{\text{KL}}(\mathcal{N}(\mu_1, \sigma_1) \parallel \mathcal{N}(\mu_2, \sigma_2)).$$

Solution:

3 Optimization [30pts]

Suppose we want to minimize the function:

$$F(x, y) = \frac{10x^2 + y^2}{2}.$$

The actual minimum is $F = 0$ at $(x^*, y^*) = (0, 0)$. Solve the following questions in *vector* notation.

1. [10 points]

Give the expression of the gradient vector ∇F at point (x, y) .

Solution:

2. [10 points] Let the initial point be $(x_0, y_0) = (1, 1)$. Perform gradient descent with step size $s = 0.5$ for two iterations. For each iteration, explicitly show:

- the gradient computation, and
- the updated solution.

Based on your results, discuss whether the resulting solution sequence will converge to the optimal solution.

Solution:

3. [10 points] Again, let $(x_0, y_0) = (1, 1)$. Perform gradient descent with step size $s = 0.1$ for two iterations. As before, clearly demonstrate:

- the gradient computation, and
- the updated solution at each step.

Determine whether the resulting sequence is convergent.

Solution:

References

Solution:

Please mention any AI tools, people, post or blog etc. you used.